Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Lesson 1: Model Selection Techniques
Introduction to Model Selection
Reading • 5 min
Bayesian Model Selection
Reading • 30 min
Bayesian Model Selection Readings
Reading • 1h

Lesson 2: Cross Validation
Introduction to Cross-Validation
Reading • 25 min
K-Fold Cross-Validation
Reading • 2 min
Leave-One-Out Cross-Validation (LOOCV)
Reading • 10 min

Lesson 3: Introduction to Optimization Techniques
Introduction to Optimization Techniques
Reading • 13 min
Optimization Using Gradient Descent
Reading • 40 min
Gradient Descent with Momentum
Reading • 45 min
Stochastic Gradient Descent (SGD)
Reading • 40 min
Constrained Optimization
Reading • 16 min
Lagrange Multipliers
Reading • 8 min
Convex Optimization
Reading • 8 min
Linear Programming
Reading • 15 min
Quadratic Programming
Reading • 10 min
Check Your Knowledge: Optimization Techniques
Graded Assignment • 15 min

Bayesian Model Selection Readings

0:16 / 0:52 2x

Read this article on Bayesian Model Selection to explore the fundamentals of matrix decomposition, including techniques such as Singular Value Decomposition (SVD) and LU decomposition. The reading also emphasizes the objective of model selection: finding the simplest model that explains the data reasonably well.

Explore Bayesian Model Selection by MIT Ton Minka, an interactive demo on statistical learning, to gain insights into how models learn from data. This resource offers visualizations and hands-on examples that illustrate key concepts in model selection, optimization, and generalization.

Read Bayesian Model Choice to deepen your understanding of Bayesian model choice and Bayesian model averaging. It explores how Bayesian methods evaluate model performance, balance complexity, and incorporate uncertainty into predictions.

Go to next item ✓ Completed

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